

A NOTE ON $\varepsilon|\log \varepsilon|$ -RATE FOR A FEYNMAN–KAC TYPE FUNCTIONAL VIA BLOCH–FLOQUET LOCAL LIMITS

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ABSTRACT. We prove, under periodic Lipschitz potential and Lipschitz terminal data, an $\varepsilon|\log \varepsilon|$ quantitative asymptotic for a rescaled one-dimensional Feynman–Kac functional. The limiting constant is described by a variational formula involving the Legendre transform of the principal eigenvalue of a tilted periodic Schrödinger operator. The proof is entirely semigroup-theoretic. We first extract the tilted principal eigenvalue on the torus via Krein–Rutman theory, then perform a ground-state transform which turns the Schrödinger evolution into a genuine periodic diffusion, and finally analyze the resulting kernel through a Bloch–Floquet decomposition and a band-edge expansion of the principal spectral branch. This yields a two-sided ballistic local-limit bound for the transformed kernel, which is transferred back to the original Schrödinger kernel and combined with a Laplace argument to obtain the stated $\varepsilon|\log \varepsilon|$ rate.

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Date: March 14, 2026.

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1. INTRODUCTION

The object studied in this paper is the small- ε asymptotic behavior of the periodic Feynman–Kac functional

$$c(\varepsilon) = \varepsilon \log \mathbb{E}_0 \left[\exp \left(-\frac{g(\varepsilon W_{2/\varepsilon})}{2\varepsilon} - \int_0^{2/\varepsilon} V(W_s) ds \right) \right].$$

The scaling is singular in two distinct ways: the observation time is of order ε^{-1} , while the terminal weight is of Laplace type and probes the macroscopic endpoint $\varepsilon W_{2/\varepsilon}$. In this regime one expects an effective variational law, but a quantitative asymptotic requires substantially more than identifying the exponential rate. The main theorem of the paper, stated precisely in Theorem 2.1, shows that

$$|c(\varepsilon) - c_0| \leq C \varepsilon |\log \varepsilon|.$$

At the exponential scale, the problem is governed by the effective phase

$$F(x) := \frac{g(x)}{2} + 2L\left(\frac{x}{2}\right),$$

where L is the Legendre transform of the tilted principal eigenvalue H . Thus the limit

$$c_0 = - \inf_{x \in \mathbb{R}} F(x)$$

is conceptually natural: $g(x)/2$ is the terminal cost and $2L(x/2)$ is the effective transport cost for reaching the macroscopic point x over macroscopic time 2. What is genuinely subtle is the passage from this qualitative variational description to a quantitative logarithmic bound. In the present periodic setting, large-deviation information alone is not sufficient. One needs a pointwise long-time asymptotic for the Schrödinger kernel along ballistic rays.

The analytic core of the paper is therefore a two-sided estimate of the form

$$K(t; 0, vt + r) \asymp t^{-1/2} e^{-tL(v)}$$

for v in compact sets and r in bounded intervals, where K denotes the kernel of the periodic Schrödinger semigroup associated with

$$\mathcal{L} = \frac{1}{2} \partial_{xx} - V(x).$$

This estimate may be viewed as a local central-limit statement for a non-selfadjoint periodic semigroup after the correct exponential tilt has been extracted. Once such a ballistic local bound is available, the original Feynman–Kac functional reduces to a one-dimensional Laplace integral, and the $\varepsilon|\log \varepsilon|$ rate follows from the low-regularity geometry of the effective phase F .

Our route to this estimate is entirely spectral and semigroup-theoretic. We first pass to the tilted torus semigroup and identify its principal eigenvalue $H(p)$ by a Krein–Rutman argument. We then perform a Doob transform with the associated positive eigenfunction, thereby replacing the Schrödinger evolution by a genuine one-dimensional diffusion with periodic drift. For this transformed diffusion we use a Bloch–Floquet decomposition of the generator, isolate the principal spectral branch, and analyze its expansion near the band edge $k = 0$. In this way the ballistic speed is recovered as $H'(p)$, while the $t^{-1/2}$ prefactor is controlled by the quadratic curvature of the principal Bloch branch. The final theorem is thus obtained by linking three layers of the problem — principal spectral data, ballistic local kernel asymptotics, and Laplace-type endpoint analysis — in a single quantitative argument.

The paper is deliberately restricted to the one-dimensional periodic setting. This is the simplest framework in which the full mechanism can be seen cleanly: the effective Hamiltonian is exact, the ground-state transform is explicit, the Bloch fibers are accessible, and the logarithmic loss in the final rate can be isolated sharply. In this sense, the note is not merely another homogenization statement; rather, it identifies a setting in which the microscopic periodic spectral geometry is still visible at the level of a quantitative macroscopic asymptotic.

1.1. Relation to the literature. The background begins with the Feynman–Kac representation for Schrödinger semigroups and parabolic equations; see, for example, [8, 18]. In the present problem, this representation converts the probabilistic functional into a long-time problem for a periodic Schrödinger operator. The periodicity then leads naturally to tilted torus operators and to principal spectral data. In particular, the map $p \mapsto H(p)$ appearing here is not introduced formally: it is the exact logarithmic growth rate of the tilted periodic semigroup.

From the viewpoint of homogenization, this places the problem in the general theory of effective Hamiltonians for periodic media. The foundational framework goes back to Bensoussan, Lions, and Papanicolaou [1]. For periodic Hamilton–Jacobi equations, Evans [7] gave a decisive viscosity-solution formulation of periodic homogenization. In closely related first-order mechanical and eikonal settings, Concorde obtained variational formulas for the effective Hamiltonian and explicit descriptions in the eikonal

regime [5, 6]. The present note belongs to this circle of ideas, but its emphasis is different: rather than stopping at the identification of the effective Hamiltonian, we exploit the tilted principal eigenvalue together with its Legendre dual to obtain a quantitative asymptotic for a concrete periodic Feynman–Kac functional.

A second line of background concerns periodic diffusions and their long-time asymptotics. For periodic diffusions, central-limit behavior and effective diffusivity are classical; see Bhattacharya [2]. What is needed here, however, is more delicate: a *two-sided local* limit estimate along ballistic rays, uniform on compact velocity sets. This is precisely the context in which Bloch–Floquet theory becomes natural. The general direct-integral framework is standard in periodic operator theory; see Kuchment [11]. In homogenization, Bloch variables were used systematically by Conca and Vanninathan [4]. A particularly close analogue in spirit is the work of Kotani and Sunada [10], where long-time off-diagonal heat-kernel asymptotics on abelian coverings are derived from perturbative analysis of twisted spectral data. The argument of the present paper imports this philosophy into the periodic diffusion obtained from the Doob transform of the Schrödinger semigroup.

It is also important to distinguish the present result from the general quantitative theory of homogenization rates for Hamilton–Jacobi equations. For first-order periodic Hamilton–Jacobi equations, the quantitative theory was initiated by Capuzzo-Dolcetta and Ishii [3]. In convex settings, sharper $O(\varepsilon)$ rates are known in important one-dimensional or structurally favorable regimes; see, for instance, [12, 19]. For periodic *viscous* Hamilton–Jacobi equations, Qian, Sprekeler, Tran, and Yu proved an $O(\sqrt{\varepsilon})$ rate in a broad general class and showed its optimality there [15]. Our theorem is not a competitor to these global PDE estimates. It addresses a much more rigid object — a single scalar periodic Feynman–Kac functional associated with a quadratic Hamiltonian and a periodic Schrödinger semigroup. Because of this extra structure, one can descend from effective Hamiltonians to ballistic local kernel asymptotics and obtain a finer logarithmic control.

For this reason, the most accurate positioning of the paper is not as a general homogenization paper, but as a meeting point of three literatures: effective Hamiltonians in periodic media, Bloch–Floquet analysis of periodic operators, and sharp long-time asymptotics for periodic heat-type kernels. The new point is that, for the present Feynman–Kac functional, these three viewpoints fit together cleanly enough to yield an explicit quantitative rate at the level of the logarithm.

1.2. Contributions and significance. The main contributions of the paper may be summarized as follows.

- (i) *A semigroup-level identification of the effective Hamiltonian.* We identify the relevant effective Hamiltonian directly as the principal eigenvalue of the tilted periodic operator

$$\mathcal{L}_p = \frac{1}{2}(\partial_y + p)^2 - V(y).$$

This places the problem in an exact spectral framework. In particular, the exponential scale is not recovered by a posteriori formal analogy with homogenization theory; it is built into the periodic Schrödinger semigroup itself.

- (ii) *A two-sided ballistic local-limit estimate via ground-state and Bloch analysis.* After the Doob transform, the problem becomes one of long-time asymptotics for a genuine periodic diffusion with drift. The key step is then a Bloch–Floquet analysis of the transformed generator, yielding a principal spectral branch whose first derivative gives the ballistic speed and whose quadratic curvature controls the local Gaussian profile. Conceptually, this shows that what survives beyond the large-deviation exponent is the band-edge geometry of the principal Bloch branch.
- (iii) *A quantitative $\varepsilon|\log \varepsilon|$ rate under only Lipschitz terminal data.* Combining the ballistic kernel estimate with a Laplace argument yields the rate

$$|c(\varepsilon) - c_0| \leq C \varepsilon |\log \varepsilon|.$$

The logarithmic loss is structural at this regularity level: if the effective phase $F(x) = \frac{1}{2}g(x) + 2L(x/2)$ has a sharp minimizer, then the Laplace integral carries a genuine logarithmic correction. Thus the theorem captures the natural worst-case behavior compatible with Lipschitz data.

At a higher level, the paper clarifies a point that is often hidden in qualitative homogenization statements: to pass from an effective Hamiltonian to a quantitative asymptotic, one needs more than the cell problem. One needs access to the long-time *local* structure of the underlying periodic semigroup. In the present setting, that local structure is encoded by the principal twisted spectral branch, and the final $\varepsilon|\log \varepsilon|$ estimate is the macroscopic shadow of that spectral fact.

1.3. Organization of the paper. The remainder of the paper is organized as follows. Section 2 introduces the functional under study, the tilted effective Hamiltonian, and the main theorem. Section 3 recalls the Feynman–Kac representation and the periodic Schrödinger kernel. Section 4 analyzes the tilted torus semigroup and defines the effective Hamiltonian H and its Legendre transform L . Section 5 performs the ground-state transform and introduces the associated periodic diffusion. Section 6 develops the Bloch–Floquet analysis and proves the local-limit estimate along ballistic rays. Section 7 transfers this estimate back to the original Schrödinger kernel. Finally, Section 8 combines the ballistic kernel asymptotics with a Laplace argument and proves Theorem 2.1.

2. SETUP AND MAIN STATEMENT

2.1. Assumptions. Throughout:

- $W = (W_t)_{t \geq 0}$ is a one-dimensional standard Brownian motion with $W_0 = 0$.
- The potential $V : \mathbb{R} \rightarrow \mathbb{R}$ is 1-periodic and Lipschitz:

$$V(x+1) = V(x), \quad V \in \text{Lip}(\mathbb{R}).$$

In particular $V \in L^\infty(\mathbb{R})$ and is continuous.

- The function $g \in W^{1,\infty}(\mathbb{R})$, hence g is bounded and globally Lipschitz.

2.2. The object of interest. For $\varepsilon > 0$, define

$$(1) \quad c(\varepsilon) := \varepsilon \log \mathbb{E}_0 \left[\exp \left(-\frac{g(\varepsilon W_{2/\varepsilon})}{2\varepsilon} - \int_0^{2/\varepsilon} V(W_s) ds \right) \right].$$

2.3. Effective Hamiltonian and the limiting constant. We will define an “effective Hamiltonian” $H : \mathbb{R} \rightarrow \mathbb{R}$ (principal eigenvalue of a tilted torus operator), and its Legendre transform L :

$$(2) \quad L(v) := \sup_{p \in \mathbb{R}} \{pv - H(p)\}.$$

Then we define

$$(3) \quad c_0 := -\inf_{x \in \mathbb{R}} \left\{ \frac{g(x)}{2} + 2L\left(\frac{x}{2}\right) \right\}.$$

2.4. Main Result.

Theorem 2.1. *Under Assumptions 2.1, there exist constants $C > 0$ and $\varepsilon_0 \in (0, 1)$ such that for all $\varepsilon \in (0, \varepsilon_0)$,*

$$(4) \quad |c(\varepsilon) - c_0| \leq C \varepsilon |\log \varepsilon|.$$

Remark 2.2. The rate $\varepsilon |\log \varepsilon|$ is the natural “worst-case” rate under only Lipschitz information: if the minimizer of $x \mapsto \frac{1}{2}g(x) + 2L(x/2)$ is sharp (e.g. a kink), Laplace-type integrals can contribute a $\log \varepsilon$ correction.

Remark 2.3. Under some additional interesting smooth nondegeneracy assumptions one typically improves to $O(\varepsilon)$, however we don’t expand this part in this note.

3. FEYNMAN–KAC REPRESENTATION AND A SCHRÖDINGER KERNEL

Set

$$(5) \quad T := \frac{2}{\varepsilon}, \quad f_\varepsilon(x) := \exp \left(-\frac{g(\varepsilon x)}{2\varepsilon} \right).$$

Define the Schrödinger-type operator on $\mathbb{R}$:

$$(6) \quad \mathcal{L} := \frac{1}{2} \partial_{xx} - V(x).$$

Lemma 3.1 (Feynman–Kac [8, Chap. 7], [18, Chap. 2]). *For bounded measurable f ,*

$$(e^{t\mathcal{L}}f)(x) = \mathbb{E}_x \left[\exp \left(-\int_0^t V(W_s) ds \right) f(W_t) \right].$$

In particular, with $T = 2/\varepsilon$,

$$(7) \quad u_\varepsilon(T, 0) := \mathbb{E}_0 \left[e^{-\int_0^T V(W_s) ds} f_\varepsilon(W_T) \right] = (e^{T\mathcal{L}}f_\varepsilon)(0),$$

and $c(\varepsilon) = \varepsilon \log u_\varepsilon(T, 0)$.

Lemma 3.2 (Kernel representation). *There exists a jointly continuous kernel $K(t; x, y) \geq 0$ such that for bounded measurable f ,*

$$(8) \quad (e^{t\mathcal{L}}f)(x) = \int_{\mathbb{R}} K(t; x, y) f(y) dy,$$

and one has the Brownian-bridge formula

$$(9) \quad K(t; x, y) = p_t(y - x) \mathbb{E}_{x,y}^{(t)} \left[\exp \left(- \int_0^t V(W_s) ds \right) \right],$$

where p_t is the Gaussian density and $\mathbb{E}_{x,y}^{(t)}$ denotes expectation under the Brownian bridge from x to y in time t .

Proof. For bounded measurable f , Feynman–Kac gives

$$(e^{t\mathcal{L}}f)(x) = \mathbb{E}_x \left[\exp \left(- \int_0^t V(W_s) ds \right) f(W_t) \right].$$

Condition on $W_t = y$. Since W_t has density $p_t(y - x)$ and the conditional law of $(W_s)_{0 \leq s \leq t}$ given $W_t = y$ is the Brownian bridge from x to y in time t , we obtain

$$(e^{t\mathcal{L}}f)(x) = \int_{\mathbb{R}} p_t(y - x) \mathbb{E}_{x,y}^{(t)} \left[\exp \left(- \int_0^t V(W_s) ds \right) \right] f(y) dy.$$

Define

$$\widehat{K}(t; x, y) := p_t(y - x) \mathbb{E}_{x,y}^{(t)} \left[\exp \left(- \int_0^t V(W_s) ds \right) \right].$$

Then the above identity shows that $\widehat{K}$ represents the semigroup as an integral operator.

We claim that $\widehat{K}$ is jointly continuous in (x, y) for each fixed $t > 0$. Indeed, for fixed $t > 0$, the Brownian bridge laws on $C([0, t]; \mathbb{R})$ depend weakly continuously on the endpoints (x, y) . Since V is bounded and continuous, the path functional

$$\omega \mapsto \exp \left(- \int_0^t V(\omega_s) ds \right)$$

is bounded and continuous on $C([0, t]; \mathbb{R})$ equipped with the uniform topology. Therefore

$$(x, y) \mapsto \mathbb{E}_{x,y}^{(t)} \left[\exp \left(- \int_0^t V(W_s) ds \right) \right]$$

is continuous, and hence so is $\widehat{K}(t; x, y)$.

On the other hand, the existence of a jointly continuous kernel in (8) is a standard consequence of parabolic regularity for

$$\partial_t u = \frac{1}{2} \partial_{xx} u - Vu$$

with bounded continuous potential V . Hence there exists a jointly continuous kernel K yielding (8).

Therefore, for every bounded measurable f and every $x \in \mathbb{R}$,

$$\int_{\mathbb{R}} K(t; x, y) f(y) dy = \int_{\mathbb{R}} \widehat{K}(t; x, y) f(y) dy.$$

So for each fixed (t, x) , the two kernels agree for a.e. y . Since both $K(t; x, y)$ and $\widehat{K}(t; x, y)$ are continuous in y , they agree for all y . Thus $K = \widehat{K}$ everywhere, which is exactly (9). $\square$

Consequently,

$$(10) \quad u_\varepsilon(T, 0) = \int_{\mathbb{R}} K(T; 0, y) f_\varepsilon(y) dy.$$

4. THE TILTED TORUS SEMIGROUP AND THE EFFECTIVE HAMILTONIAN

4.1. The tilted moment generating function. For $p \in \mathbb{R}$ and $t > 0$ define

$$(11) \quad \Psi_t(p) := \mathbb{E}_0 \left[\exp \left(pW_t - \int_0^t V(W_s) ds \right) \right] = \int_{\mathbb{R}} e^{py} K(t; 0, y) dy.$$

4.2. Torus reduction. Let $\mathbb{T} := \mathbb{R}/\mathbb{Z}$ and define the torus operator

$$(12) \quad \mathcal{L}_p := \frac{1}{2}(\partial_y + p)^2 - V(y) = \frac{1}{2}\partial_{yy} + p\partial_y + \frac{1}{2}p^2 - V(y) \quad \text{on } \mathbb{T}.$$

Lemma 4.1 (Torus semigroup identity). *Let $S_t^{(p)} := e^{t\mathcal{L}_p}$ be the semigroup generated by $\mathcal{L}_p$ on $C(\mathbb{T})$. Then for all $t > 0$,*

$$(13) \quad \Psi_t(p) = (S_t^{(p)} \mathbf{1})(0).$$

Proof. Let $U(t, x) := \mathbb{E}_x[\exp(pW_t - \int_0^t V(W_s) ds)]$. A standard localization/approximation argument extends the Feynman–Kac formula to the exponentially growing initial datum e^{px} , so U solves

$$\partial_t U = \frac{1}{2}\partial_{xx}U - V(x)U, \quad U(0, x) = e^{px}.$$

Write $U(t, x) = e^{px}v(t, x)$. Direct differentiation gives

$$\partial_t v = \frac{1}{2}\partial_{xx}v + p\partial_x v + \frac{1}{2}p^2v - V(x)v = \mathcal{L}_p v, \quad v(0, x) \equiv 1.$$

Since V is 1-periodic and the initial condition is 1-periodic, uniqueness of solutions implies $v(t, \cdot)$ remains 1-periodic, so $v(t, \cdot)$ can be viewed on $\mathbb{T}$ and $v(t, 0) = (e^{t\mathcal{L}_p} \mathbf{1})(0)$. Finally $U(t, 0) = v(t, 0)$ gives (13). $\square$

4.3. Krein–Rutman principal eigenvalue. For fixed $t > 0$, $S_t^{(p)}$ has a strictly positive continuous kernel on $\mathbb{T}$ (uniform ellipticity on a compact manifold plus a strictly positive Feynman–Kac weight), hence it is compact on $C(\mathbb{T})$ and positivity improving.

Lemma 4.2 (Krein–Rutman [17, Thm. V.6.6]). *For each $p \in \mathbb{R}$ there exist:*

- a real number $H(p)$ (the principal eigenvalue / spectral bound),
- a strictly positive eigenfunction $r_p \in C(\mathbb{T})$, $r_p > 0$,
- a strictly positive eigenfunctional ℓ_p (equivalently a positive Borel probability measure after normalization),

such that for all $t > 0$,

$$S_t^{(p)} r_p = e^{tH(p)} r_p, \quad (S_t^{(p)})^* \ell_p = e^{tH(p)} \ell_p,$$

and the eigenvalue $e^{tH(p)}$ is algebraically simple and strictly dominates the rest of the spectrum of $S_t^{(p)}$.

For definiteness, we henceforth fix the positive normalization

$$r_p(0) = 1 \quad (p \in \mathbb{R}).$$

Since the principal eigenspace is one-dimensional and $r_p > 0$, this normalization depends continuously (indeed locally analytically) on p .

Lemma 4.3 (Spectral-gap decomposition on $C(\mathbb{T})$). *Fix $p \in \mathbb{R}$. There exist constants $\gamma(p) > 0$ and $C(p) \geq 1$ and a rank-one projection*

$$(P_p \varphi)(x) := r_p(x) \frac{\langle \ell_p, \varphi \rangle}{\langle \ell_p, r_p \rangle} \quad (\varphi \in C(\mathbb{T})),$$

such that for all $t \geq 0$,

$$(14) \quad S_t^{(p)} = e^{tH(p)} P_p + R_t^{(p)}, \quad \left\| R_t^{(p)} \right\|_{C(\mathbb{T}) \rightarrow C(\mathbb{T})} \leq C(p) e^{(H(p) - \gamma(p))t}.$$

Proof. Apply the Riesz projection associated to the isolated simple eigenvalue $e^{H(p)}$ of $S_1^{(p)}$ and use compactness to obtain a spectral gap. Then extend to all $t \geq 0$ by the semigroup property. $\square$

Corollary 4.4. *For each fixed $p \in \mathbb{R}$ there exist $t_0(p) \geq 1$ and constants $0 < c(p) \leq C(p) < \infty$ such that for all $t \geq t_0(p)$,*

$$(15) \quad c(p) e^{tH(p)} \leq \Psi_t(p) \leq C(p) e^{tH(p)}.$$

Proof. By Lemma 4.3 and Lemma 4.1,

$$\Psi_t(p) = (S_t^{(p)} 1)(0) = e^{tH(p)} (P_p 1)(0) + (R_t^{(p)} 1)(0),$$

and $(P_p 1)(0) > 0$. The remainder term is exponentially small relative to $e^{tH(p)}$ for large t . $\square$

Corollary 4.5 (Uniform two-sided bound on compact p -sets). *Let $P \subset \mathbb{R}$ be compact. Then there exist $t_P^\Psi \geq 1$ and constants $0 < c_P^\Psi \leq C_P^\Psi < \infty$ such that for all $p \in P$ and all $t \geq t_P^\Psi$,*

$$c_P^\Psi e^{tH(p)} \leq \Psi_t(p) \leq C_P^\Psi e^{tH(p)}.$$

Proof. For each p , Lemma 4.3 gives

$$\Psi_t(p) = e^{tH(p)} (P_p 1)(0) + (R_t^{(p)} 1)(0).$$

For fixed $t = 1$, the torus semigroup $S_1^{(p)}$ admits a jointly continuous kernel $K_{p,1}^\mathbb{T}(x, y)$ on $\mathbb{T} \times \mathbb{T}$, and $(p, x, y) \mapsto K_{p,1}^\mathbb{T}(x, y)$ is continuous on compact sets. Hence, if $p_n \rightarrow p$,

$$\|S_1^{(p_n)} - S_1^{(p)}\|_{C(\mathbb{T}) \rightarrow C(\mathbb{T})} \leq \sup_{x \in \mathbb{T}} \int_{\mathbb{T}} |K_{p_n,1}^\mathbb{T}(x, y) - K_{p,1}^\mathbb{T}(x, y)| dy \rightarrow 0.$$

Therefore $p \mapsto S_1^{(p)}$ is continuous in operator norm on $C(\mathbb{T})$. Therefore, by standard perturbation theory for isolated simple eigenvalues of compact operators (e.g. [9, Chap. VII]), the eigenprojection P_p and the spectral-gap data in (14) depend continuously on p in operator norm. Hence on compact P there exist constants $\gamma_P > 0$ and $C_P^{\text{gap}} \geq 1$ such that

$$\|(R_t^{(p)} 1)\|_\infty \leq C_P^{\text{gap}} e^{(H(p) - \gamma_P)t} \quad \text{for all } p \in P, t \geq 0.$$

Moreover $a(p) := (P_p 1)(0) > 0$ and $p \mapsto a(p)$ is continuous, so $a_* := \min_{p \in P} a(p) > 0$ and $a^* := \max_{p \in P} a(p) < \infty$.

Choose t_P^Ψ large so that

$$C_P^{\text{gap}} e^{-\gamma_P t_P^\Psi} \leq \min\{a_*/2, 1\}.$$

Then for all $t \geq t_P^\Psi$ and $p \in P$,

$$\Psi_t(p) \geq e^{tH(p)} \left(a_* - C_P^{\text{gap}} e^{-\gamma_P t} \right) \geq \frac{a_*}{2} e^{tH(p)},$$

and similarly

$$\Psi_t(p) \leq e^{tH(p)} \left(a^* + C_P^{\text{gap}} e^{-\gamma_P t} \right) \leq (a^* + 1) e^{tH(p)}.$$

Set $c_P^\Psi := a_*/2$ and $C_P^\Psi := a^* + 1$.

□

4.4. Basic properties of H and L .

Lemma 4.6 (Quadratic bounds for H). *For all $p \in \mathbb{R}$,*

$$(16) \quad \frac{p^2}{2} - \|V\|_\infty \leq H(p) \leq \frac{p^2}{2} + \|V\|_\infty.$$

Proof. From (11) and boundedness of V ,

$$e^{-\|V\|_\infty t} \mathbb{E}[e^{pW_t}] \leq \Psi_t(p) \leq e^{\|V\|_\infty t} \mathbb{E}[e^{pW_t}],$$

and $\mathbb{E}[e^{pW_t}] = e^{p^2 t/2}$. Divide by t and let $t \rightarrow \infty$ using $\lim_{t \rightarrow \infty} \frac{1}{t} \log \Psi_t(p) = H(p)$, which follows from Corollary 4.4. □

Lemma 4.7 (Analyticity of H [9, Chap. VII, §1]). *The map $p \mapsto H(p)$ is real-analytic on $\mathbb{R}$. Moreover, the eigenpair $(H(p), r_p)$ can be chosen locally analytic in p in suitable Banach/Hilbert topologies.*

Proof. On $L^2(\mathbb{T})$ with domain $H^2(\mathbb{T})$, the family

$$\mathcal{L}_p = \frac{1}{2} \partial_{yy} + p \partial_y + \frac{1}{2} p^2 - V$$

is a Kato type-(A) analytic family: it depends polynomially on p and has p -independent domain.

For each fixed p and each $t > 0$, the semigroup $e^{t\mathcal{L}_p}$ on $L^2(\mathbb{T})$ has a strictly positive continuous kernel on $\mathbb{T}$ (by the same Feynman–Kac argument as on $C(\mathbb{T})$), hence it is compact and positivity improving. Therefore its principal eigenvalue is algebraically simple. Kato’s analytic perturbation theory then yields that $p \mapsto H(p)$ is real-analytic

on $\mathbb{R}$, and the corresponding eigenpair may be chosen locally analytic in suitable Banach/Hilbert topologies. $\square$

Lemma 4.8 (Convexity of $\log \Psi_t$ and H). *For each $t > 0$, the map $p \mapsto \log \Psi_t(p)$ is convex on $\mathbb{R}$. Consequently,*

$$H(p) = \lim_{t \rightarrow \infty} \frac{1}{t} \log \Psi_t(p)$$

is convex on $\mathbb{R}$.

Proof. Fix $t > 0$. For $p_0, p_1 \in \mathbb{R}$ and $\theta \in [0, 1]$, Hölder's inequality gives

$$\Psi_t(\theta p_1 + (1-\theta)p_0) = \mathbb{E}_0 \left[\exp \left((\theta p_1 + (1-\theta)p_0) W_t - \int_0^t V(W_s) ds \right) \right] \leq \Psi_t(p_1)^\theta \Psi_t(p_0)^{1-\theta}.$$

Taking log yields convexity of $\log \Psi_t$. Dividing by t and letting $t \rightarrow \infty$ gives convexity of H . $\square$

Define L by (2). Standard convex analysis (e.g. [16, Chap. 12]) implies L is convex and has quadratic growth: from (16) we immediately get

$$(17) \quad \frac{v^2}{2} - \|V\|_\infty \leq L(v) \leq \frac{v^2}{2} + \|V\|_\infty, \quad v \in \mathbb{R}.$$

5. GROUND-STATE TRANSFORM AND THE PERIODIC DIFFUSION

5.1. Regularity of the principal eigenfunction. By Lemma 4.7, the principal eigenfunction may be chosen in $H^2(\mathbb{T})$. Since we are in one dimension,

$$H^2(\mathbb{T}) \hookrightarrow C^1(\mathbb{T}).$$

Moreover, the eigenvalue equation

$$\frac{1}{2} r_p'' + p r_p' + \left(\frac{1}{2} p^2 - V - H(p) \right) r_p = 0$$

shows that $r_p'' \in C(\mathbb{T})$ because V is continuous and $r_p, r_p' \in C(\mathbb{T})$. Hence in fact $r_p \in C^2(\mathbb{T})$.

In particular, r_p is bounded away from 0 and ∞ on $\mathbb{T}$.

5.2. The ground state on $\mathbb{R}$. Define the lifted positive eigenfunction on $\mathbb{R}$:

$$(18) \quad h_p(x) := e^{px} r_p(x \bmod 1).$$

Lemma 5.1 (Lifted eigenfunction). *On $\mathbb{R}$,*

$$(19) \quad \mathcal{L} h_p = H(p) h_p.$$

Proof. View r_p as a 1-periodic C^2 function on $\mathbb{R}$. Then

$$h_p' = e^{px} (p r_p + r_p'), \quad h_p'' = e^{px} (p^2 r_p + 2p r_p' + r_p'').$$

Hence

$$\mathcal{L} h_p = \frac{1}{2} h_p'' - V h_p = e^{px} \left(\frac{1}{2} (r_p'' + 2p r_p' + p^2 r_p) - V r_p \right) = e^{px} (\mathcal{L}_p r_p) = H(p) h_p.$$

$\square$

5.3. Doob h -transform. Define for bounded measurable f :

$$(20) \quad (\tilde{P}_t^{(p)} f)(x) := e^{-tH(p)} h_p(x)^{-1} \mathbb{E}_x \left[\exp \left(- \int_0^t V(W_s) ds \right) h_p(W_t) f(W_t) \right], \quad t \geq 0.$$

Lemma 5.2 (Ground-state / Doob transform [14, Sec. 4.1]). *For each $p \in \mathbb{R}$, $(\tilde{P}_t^{(p)})_{t \geq 0}$ is a Markov semigroup on bounded measurable functions: it preserves positivity, satisfies $\tilde{P}_t^{(p)} \mathbf{1} = \mathbf{1}$, and has the semigroup property.*

Proof. Positivity is immediate from the definition and the fact that $h_p > 0$. Since

$$h_p(x) = e^{px} r_p(x \bmod 1),$$

with $r_p \in C^2(\mathbb{T})$ strictly positive, the function

$$b_p(x) := \frac{h'_p(x)}{h_p(x)}$$

is bounded on $\mathbb{R}$.

Using (19), Itô's formula gives

$$d \left(e^{-tH(p)} \exp \left(- \int_0^t V(W_s) ds \right) h_p(W_t) \right) = e^{-tH(p)} \exp \left(- \int_0^t V(W_s) ds \right) h_p(W_t) b_p(W_t) dW_t.$$

Hence

$$M_t := e^{-tH(p)} \exp \left(- \int_0^t V(W_s) ds \right) h_p(W_t) = h_p(x) \exp \left(\int_0^t b_p(W_s) dW_s - \frac{1}{2} \int_0^t |b_p(W_s)|^2 ds \right).$$

Since b_p is bounded, Novikov's condition holds on every finite time interval, so $(M_t)_{t \geq 0}$ is a true martingale under $\mathbb{P}_x$.

Therefore

$$\tilde{P}_t^{(p)} \mathbf{1} = \mathbf{1}.$$

The semigroup property follows from the Markov property of Brownian motion together with the multiplicative structure of the Feynman–Kac weight. $\square$

Lemma 5.3 (Generator and SDE). *The generator of $\tilde{P}_t^{(p)}$ is*

$$(21) \quad \tilde{L}_p f = \frac{1}{2} f'' + b_p(x) f', \quad b_p(x) := \frac{h'_p(x)}{h_p(x)} = p + \frac{r'_p(x \bmod 1)}{r_p(x \bmod 1)}.$$

In particular b_p is 1-periodic and Lipschitz, hence the diffusion

$$(22) \quad dX_t^{(p)} = b_p(X_t^{(p)}) dt + dB_t$$

is well-posed and has transition densities.

Proof. For $f \in C_b^2(\mathbb{R})$, define

$$N_t^f := e^{-tH(p)} \exp \left(- \int_0^t V(W_s) ds \right) h_p(W_t) f(W_t).$$

Since $h_p \in C^2(\mathbb{R})$ and $\mathcal{L}h_p = H(p)h_p$, Itô's formula gives

$$dN_t^f = e^{-tH(p)} \exp\left(-\int_0^t V(W_s) ds\right) (\mathcal{L}(h_p f) - H(p)h_p f)(W_t) dt + dM_t,$$

where (M_t) is a local martingale. Using

$$\mathcal{L}(h_p f) - H(p)h_p f = h_p \left(\frac{1}{2} f'' + \frac{h_p'}{h_p} f' \right),$$

we obtain

$$dN_t^f = e^{-tH(p)} \exp\left(-\int_0^t V(W_s) ds\right) h_p(W_t) \tilde{L}_p f(W_t) dt + dM_t,$$

with

$$\tilde{L}_p f = \frac{1}{2} f'' + \frac{h_p'}{h_p} f' = \frac{1}{2} f'' + b_p f'.$$

Let $\tau_n := \inf\{s \geq 0 : |W_s| \geq n\}$. Integrating the above identity up to time $t \wedge \tau_n$ and taking expectations, the stopped local martingale term has zero expectation. Since b_p, f, f', f'' are bounded and

$$\mathbb{E}_x \left[e^{-sH(p)} \exp\left(-\int_0^s V(W_u) du\right) h_p(W_s) \right] = h_p(x) \quad (0 \leq s \leq t),$$

dominated convergence lets us pass to the limit $n \rightarrow \infty$. Hence

$$(\tilde{P}_t^{(p)} f)(x) = f(x) + \int_0^t (\tilde{P}_s^{(p)} \tilde{L}_p f)(x) ds, \quad f \in C_b^2(\mathbb{R}).$$

Hence the generator of $(\tilde{P}_t^{(p)})_{t \geq 0}$ on $C_b^2(\mathbb{R})$ is (21).

Since b_p is Lipschitz and 1-periodic, the SDE (22) is well posed and its martingale problem for

$$\frac{1}{2} \partial_{xx} + b_p(x) \partial_x$$

is unique. Therefore the Markov semigroup $(\tilde{P}_t^{(p)})_{t \geq 0}$ coincides with the transition semigroup of $X^{(p)}$. In particular,

$$(\tilde{P}_t^{(p)} f)(x) = \int_{\mathbb{R}} \tilde{p}_p(t, x, y) f(y) dy,$$

where $\tilde{p}_p(t, x, y)$ is the strictly positive jointly continuous transition density of $X^{(p)}$. $\square$

5.4. Kernel identity. Let $\tilde{p}_p(t, x, y)$ denote the transition density of $X^{(p)}$ (w.r.t. Lebesgue measure).

Lemma 5.4 (Kernel identity). *For all $t > 0$ and $x, y \in \mathbb{R}$,*

$$(23) \quad K(t; x, y) = e^{tH(p)} \frac{h_p(x)}{h_p(y)} \tilde{p}_p(t, x, y).$$

Proof. Since the SDE (22) with Lipschitz drift is well posed and its martingale problem is unique, the Markov semigroup $(\tilde{P}_t^{(p)})_{t \geq 0}$ with generator (21) coincides with the transition semigroup of $X^{(p)}$. Hence

$$(\tilde{P}_t^{(p)} f)(x) = \int_{\mathbb{R}} \tilde{p}_p(t, x, y) f(y) dy.$$

For bounded measurable f , the definition (20) gives

$$e^{tH(p)} h_p(x) (\tilde{P}_t^{(p)} f)(x) = \mathbb{E}_x \left[\exp \left(- \int_0^t V(W_s) ds \right) h_p(W_t) f(W_t) \right].$$

Conditioning on $W_t = y$ and using Lemma 3.2, we obtain

$$e^{tH(p)} h_p(x) (\tilde{P}_t^{(p)} f)(x) = \int_{\mathbb{R}} K(t; x, y) h_p(y) f(y) dy.$$

Therefore, for each fixed x and every bounded measurable f ,

$$\int_{\mathbb{R}} \left(e^{tH(p)} h_p(x) \tilde{p}_p(t, x, y) - K(t; x, y) h_p(y) \right) f(y) dy = 0.$$

Hence for each fixed x , the bracketed functions agree for a.e. y . Since both sides are continuous in y , they agree for all y . Dividing by $h_p(x) h_p(y) > 0$ yields (23) for all x, y . $\square$

6. FROM THE PERIODIC DIFFUSION TO A BLOCH LOCAL LIMIT BOUND

Fix $p \in \mathbb{R}$ for this section, and abbreviate $\tilde{p} := \tilde{p}_p$ and $b := b_p$ when no confusion arises.

6.1. Invariant density and the ballistic speed m_p . Let $Y_t^{(p)} := X_t^{(p)} \bmod 1$ be the induced torus diffusion. Its generator on $\mathbb{T}$ is

$$(24) \quad \tilde{\mathcal{G}}_p f = \frac{1}{2} f'' + b_p(y) f'.$$

Since b_p is Lipschitz and the diffusion is uniformly elliptic on the compact manifold $\mathbb{T}$, $Y^{(p)}$ is strong Feller and topologically irreducible; hence it has a unique invariant density $\pi_p \in C^1(\mathbb{T})$ with $\pi_p > 0$.

Define the (law of large numbers) speed

$$(25) \quad m_p := \int_0^1 b_p(y) \pi_p(y) dy.$$

Let $\mathcal{L}_p^* = \frac{1}{2}(\partial_y - p)^2 - V(y)$ be the adjoint of $\mathcal{L}_p$ on $L^2(\mathbb{T})$. Let $l_p > 0$ solve $\mathcal{L}_p^* l_p = H(p) l_p$ and normalize

$$(26) \quad \int_0^1 l_p(y) r_p(y) dy = 1.$$

Lemma 6.1 (Invariant density). *With normalization (26), the invariant density is*

$$(27) \quad \pi_p(y) = l_p(y) r_p(y).$$

Proof. The adjoint of (24) acting on densities is

$$\tilde{\mathcal{G}}_p^* \rho = \frac{1}{2} \rho'' - (b_p \rho)'.$$

Set $\rho = l_p r_p$. Using $b_p = p + r'_p / r_p$ and expanding,

$$\tilde{\mathcal{G}}_p^*(l_p r_p) = \left(\frac{1}{2} l_p'' - p l_p' \right) r_p - l_p \left(\frac{1}{2} r_p'' + p r_p' \right).$$

Now use the eigenvalue equations

$$\mathcal{L}_p^* l_p = \left(\frac{1}{2} l_p'' - p l_p' + \frac{1}{2} p^2 l_p - V l_p \right) = H(p) l_p, \quad \mathcal{L}_p r_p = \left(\frac{1}{2} r_p'' + p r_p' + \frac{1}{2} p^2 r_p - V r_p \right) = H(p) r_p,$$

to deduce

$$\frac{1}{2} l_p'' - p l_p' = (H(p) + V - \frac{1}{2} p^2) l_p, \quad \frac{1}{2} r_p'' + p r_p' = (H(p) + V - \frac{1}{2} p^2) r_p,$$

and hence $\tilde{\mathcal{G}}_p^*(l_p r_p) = 0$. Uniqueness of the invariant density on $\mathbb{T}$ yields (27) after normalizing to integrate to 1, which is exactly (26). $\square$

Lemma 6.2 (Hellmann–Feynman identity [9, Chap. VII, §3]). *One has*

$$(28) \quad m_p = H'(p).$$

Proof. Using (25), (27), and (26),

$$(29) \quad m_p = \int_0^1 \left(p + \frac{r'_p}{r_p} \right) l_p r_p dy = p + \int_0^1 l_p r'_p dy.$$

By Lemma 4.7, H is analytic and the eigenpair is differentiable. Under normalization $\int l_p r_p = 1$, the Hellmann–Feynman formula gives

$$H'(p) = \int_0^1 l_p (\partial_p \mathcal{L}_p) r_p dy.$$

Since $\partial_p \mathcal{L}_p = \partial_y + p$, we obtain

$$H'(p) = \int_0^1 l_p (r'_p + p r_p) dy = p + \int_0^1 l_p r'_p dy.$$

Comparing with (29) yields (28). $\square$

6.2. Bloch–Floquet decomposition for the periodic diffusion generator. Consider the generator on $\mathbb{R}$:

$$(30) \quad \tilde{L}_p = \frac{1}{2} \partial_{xx} + b_p(x) \partial_x, \quad b_p(x+1) = b_p(x).$$

For $k \in [-\pi, \pi]$, define the fiber operator on $\mathbb{T}$ on complex-valued periodic functions:

$$(31) \quad \mathcal{A}_p(k) \varphi := \frac{1}{2} (\partial_y + ik)^2 \varphi + b_p(y) (\partial_y + ik) \varphi, \quad \varphi \in H^2(\mathbb{T}; \mathbb{C}).$$

From this point on, all fiber operators and fiber semigroups are understood on complex-valued spaces

$$H^2(\mathbb{T}; \mathbb{C}), \quad L^2(\mathbb{T}; \mathbb{C}), \quad C(\mathbb{T}; \mathbb{C}).$$

Proposition 6.3 (Bloch–Floquet direct integral [11, Chaps. 2–3]). *Let $\tilde{p}_p(t, x, y)$ be the transition density of the diffusion (22). Then for all $t > 0$ and $x, y \in \mathbb{R}$,*

$$(32) \quad \tilde{p}_p(t, x, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ik(x-y)} p_{p,k}(t; x \bmod 1, y \bmod 1) dk,$$

where $p_{p,k}(t; \xi, \eta)$ is the (jointly continuous) integral kernel of the fiber semigroup $e^{t\mathcal{A}_p(k)}$ on $\mathbb{T}$:

$$(e^{t\mathcal{A}_p(k)}\phi)(\xi) = \int_{\mathbb{T}} p_{p,k}(t; \xi, \eta)\phi(\eta) d\eta.$$

Remark 6.4. The Bloch variable $k \in [-\pi, \pi]$ lives in a *fundamental Brillouin zone*. This removes the “ $2\pi\mathbb{Z}$ gauge” obstruction.

Remark 6.5 (Operator-level direct integral). Equation (32) is the kernel version of the direct-integral decomposition

$$\mathcal{B} \tilde{L}_p \mathcal{B}^{-1} = \int_{[-\pi, \pi]}^{\oplus} \mathcal{A}_p(k) dk \quad \text{on} \quad L^2(\mathbb{R}; \mathbb{C}) \simeq \int_{[-\pi, \pi]}^{\oplus} L^2(\mathbb{T}; \mathbb{C}) dk,$$

where $\mathcal{B}$ is the Bloch (Zak) transform; see [11, Chaps. 2–3]. The semigroup identity

$$\mathcal{B} e^{t\tilde{L}_p} \mathcal{B}^{-1} = \int_{[-\pi, \pi]}^{\oplus} e^{t\mathcal{A}_p(k)} dk$$

then yields (32).

6.3. Principal band: spectral gap and analytic expansion. For each fixed (p, k) , $\mathcal{A}_p(k)$ is an elliptic operator on the compact manifold $\mathbb{T}$ with compact resolvent, hence it has discrete spectrum $\{\lambda_{p,n}(k)\}_{n \geq 0}$ with $\Re \lambda_{p,n}(k) \rightarrow -\infty$. Let $s(\mathcal{A}_p(k)) := \sup\{\Re \lambda : \lambda \in \sigma(\mathcal{A}_p(k))\}$ denote its spectral bound, and let $\lambda_p(k)$ denote an eigenvalue attaining this bound (the principal band).

Lemma 6.6. $\lambda_p(0) = 0$, and 0 is a simple eigenvalue of $\mathcal{A}_p(0)$. The right eigenfunction is the constant 1, and the left eigenfunction is π_p (the invariant density of $Y^{(p)}$).

Proof. $\mathcal{A}_p(0) = \tilde{\mathcal{G}}_p$ is a Markov generator on $\mathbb{T}$, hence $e^{t\mathcal{A}_p(0)}1 = 1$ and 0 is an eigenvalue.

Since $e^{t\mathcal{A}_p(0)}$ is a compact positivity-improving Markov semigroup on $C(\mathbb{T})$, the eigenvalue 1 of $e^{t\mathcal{A}_p(0)}$ is simple by Krein–Rutman. Hence 0 is a simple eigenvalue of $\mathcal{A}_p(0)$. The corresponding right eigenfunction is 1, and the left eigenfunction is the invariant density π_p . $\square$

Lemma 6.7 (Twisted spectral gap away from $k = 0$). *Fix $p \in \mathbb{R}$. For any $k_0 \in (0, \pi]$ there exists $\eta(p, k_0) > 0$ such that*

$$(33) \quad \sup_{|k| \geq k_0} \Re \lambda_p(k) = \sup_{|k| \geq k_0} s(\mathcal{A}_p(k)) \leq -\eta(p, k_0).$$

Proof. Fix $\tau > 0$ and let $X^{(p)}$ be the diffusion on $\mathbb{R}$ solving (22). For $k \in [-\pi, \pi]$, define an operator $T_{p,k}$ on $C(\mathbb{T}; \mathbb{C})$ as follows. Given $\xi \in \mathbb{T}$, choose any lift $x \in \mathbb{R}$ such that $x \bmod 1 = \xi$, and let $X^{(p),x}$ solve (22) with initial condition $X_0^{(p),x} = x$. Set

$$(34) \quad (T_{p,k}\varphi)(\xi) := \mathbb{E} \left[e^{ik(X_\tau^{(p),x} - x)} \varphi(X_\tau^{(p),x} \bmod 1) \right].$$

Because b_p is 1-periodic, if x is replaced by $x+n$ with $n \in \mathbb{Z}$, then $X_t^{(p),x+n} = X_t^{(p),x} + n$ pathwise, so the right-hand side is independent of the chosen lift.

A standard Itô/conjugation calculation shows that

$$T_{p,k} = e^{\tau \mathcal{A}_p(k)} \quad \text{on } C(\mathbb{T}; \mathbb{C}),$$

where the phase factor $e^{ik(X_t^{(p),x} - x)}$ produces the twists $(\partial_y + ik)$ in (31).

In particular, if x is any lift of ξ , then

$$(35) \quad |(T_{p,k}\varphi)(\xi)| \leq \mathbb{E}[|\varphi(X_\tau^{(p),x} \bmod 1)|] = (T_{p,0}|\varphi|)(\xi),$$

and $T_{p,0}$ is a Markov operator with strictly positive continuous kernel on $\mathbb{T}$, hence positivity improving. *Step 1:* $s(\mathcal{A}_p(k)) < 0$ for all $k \neq 0$. Assume for contradiction that $s(\mathcal{A}_p(k)) = 0$ for some $k \in (-\pi, \pi] \setminus \{0\}$. By spectral mapping for compact semigroups,

$$r(T_{p,k}) = e^{\tau s(\mathcal{A}_p(k))} = 1,$$

where $r(\cdot)$ denotes spectral radius. Since $T_{p,k}$ is compact on $C(\mathbb{T})$, there exist $\varphi \in C(\mathbb{T}) \setminus \{0\}$ and $\theta \in \mathbb{R}$ such that

$$T_{p,k}\varphi = e^{i\theta}\varphi.$$

Let $M := \max_{\xi \in \mathbb{T}} |\varphi(\xi)| > 0$ and pick ξ_0 with $|\varphi(\xi_0)| = M$. Then using (35) and the Markov property of $T_{p,0}$,

$$M = |\varphi(\xi_0)| = |T_{p,k}\varphi(\xi_0)| \leq (T_{p,0}|\varphi|)(\xi_0) \leq (T_{p,0}M)(\xi_0) = M.$$

Thus equality holds throughout. Positivity improving implies $|\varphi| \equiv M$ on $\mathbb{T}$.

Moreover, equality in the triangle inequality in (34) forces the following. Choose a lift $x_0 \in \mathbb{R}$ of ξ_0 . Then the random variable

$$e^{ik(X_\tau^{(p),x_0} - x_0)} \varphi(X_\tau^{(p),x_0} \bmod 1)$$

has constant complex argument a.s.

Since $X_\tau^{(p),x_0}$ has a strictly positive transition density on $\mathbb{R}$, the above constant-argument property holds for Lebesgue-a.e. $y \in \mathbb{R}$. Since $|\varphi| \equiv M > 0$ and the map

$$y \mapsto e^{ik(y-x_0)} \varphi(y \bmod 1)$$

is continuous, it in fact has constant argument for all $y \in \mathbb{R}$.

Hence $\varphi(y \bmod 1) = Ce^{-iky}$ for some $C \neq 0$. Because φ is 1-periodic and continuous, we must have $e^{-ik(y+1)} = e^{-iky}$, i.e. $e^{-ik} = 1$. Thus $k \in 2\pi\mathbb{Z}$, and with $k \in (-\pi, \pi]$ we get $k = 0$, contradiction. Therefore $s(\mathcal{A}_p(k)) < 0$ for all $k \neq 0$.

Step 2: uniformity on $|k| \geq k_0$. The map $k \mapsto s(\mathcal{A}_p(k))$ is upper semicontinuous (e.g. via norm-resolvent continuity of the elliptic compact-resolvent family $k \mapsto \mathcal{A}_p(k)$). On the compact set $\{k \in [-\pi, \pi] : |k| \in [k_0, \pi]\}$ it attains its maximum, which is strictly negative by Step 1. Let $\eta(p, k_0) > 0$ be minus that maximum. This yields (33). $\square$

Lemma 6.8 (Analytic expansion near $k = 0$ [9, Chap. VII, §1]). *There exists $\delta(p) > 0$ such that for $|k| < \delta(p)$ the principal band $\lambda_p(k)$ is analytic in k and admits the Taylor expansion*

$$(36) \quad \lambda_p(k) = i m_p k - \frac{1}{2} \sigma_p^2 k^2 + O(|k|^3) \quad (k \rightarrow 0),$$

where $m_p = H'(p)$ and $\sigma_p^2 > 0$.

Proof. The family $k \mapsto \mathcal{A}_p(k)$ is a Kato type-(A) analytic family on $L^2(\mathbb{T})$ with fixed domain $H^2(\mathbb{T})$, since it is polynomial in ik with bounded coefficients. At $k = 0$, $\lambda_p(0) = 0$ is simple (Lemma 6.6), hence analytic perturbation theory yields an analytic eigenvalue branch $\lambda_p(k)$ for $|k| < \delta(p)$.

The coefficient of the linear term equals im_p ; differentiate at $k = 0$ in the eigenvalue equation and use that the left eigenfunction at $k = 0$ is π_p and the right eigenfunction is 1, which yields $\lambda_p'(0) = i \int b_p \pi_p = im_p$. Identifying $m_p = H'(p)$ is Lemma 6.2.

To identify the quadratic term, let $\chi_p \in C^2(\mathbb{T})$ be the unique mean-zero solution of the Poisson equation

$$\tilde{\mathcal{G}}_p \chi_p = -(b_p - m_p), \quad \int_{\mathbb{T}} \chi_p(y) \pi_p(y) dy = 0.$$

Existence and uniqueness hold because 0 is a simple isolated eigenvalue of $\tilde{\mathcal{G}}_p$ and the right-hand side has zero π_p -mean. Applying Itô's formula to $\chi_p(Y_t^{(p)})$ gives

$$\chi_p(Y_t^{(p)}) - \chi_p(Y_0^{(p)}) = - \int_0^t (b_p(Y_s^{(p)}) - m_p) ds + \int_0^t \chi_p'(Y_s^{(p)}) dB_s.$$

Since

$$X_t^{(p)} - m_p t = \int_0^t (b_p(Y_s^{(p)}) - m_p) ds + B_t,$$

we obtain the martingale-corrector decomposition

$$X_t^{(p)} - m_p t = M_t^{(p)} + \chi_p(Y_0^{(p)}) - \chi_p(Y_t^{(p)}),$$

where

$$M_t^{(p)} := \int_0^t (1 + \chi_p'(Y_s^{(p)})) dB_s.$$

Therefore the asymptotic variance is

$$\sigma_p^2 = \int_{\mathbb{T}} |1 + \chi_p'(y)|^2 \pi_p(y) dy,$$

The identity

$$-\lambda_p''(0) = \sigma_p^2$$

is the standard identification of the second derivative of the principal Bloch eigenvalue with the effective diffusivity of the periodic diffusion; see [13, Chap. 13]. Finally, $\sigma_p^2 > 0$: if $\sigma_p^2 = 0$, then

$$1 + \chi_p'(y) = 0$$

for π_p -a.e. y , hence everywhere by continuity, which is impossible on the torus since

$$\int_{\mathbb{T}} \chi_p'(y) dy = 0.$$

This completes the proof. □

6.4. A uniformity on compact p -sets.

Proposition 6.9 (Uniform spectral data on compact p -sets). *Let $P \subset \mathbb{R}$ be compact. Then there exist constants*

$$\delta_P \in (0, 1), \quad \sigma_P > 0, \quad \eta_P > 0, \quad C_P \geq 1,$$

and $t_P \geq 1$ such that the following hold for every $p \in P$:

(a) (Uniform small- k expansion) For $|k| \leq \delta_P$,

$$\lambda_p(k) = i m_p k - \frac{1}{2} \sigma_p^2 k^2 + R_p(k), \quad |R_p(k)| \leq C_P |k|^3,$$

and $\sigma_p^2 \geq \sigma_P$.

(b) (Uniform decay away from $k = 0$) For $|k| \in [\delta_P, \pi]$,

$$s(\mathcal{A}_p(k)) = \Re \lambda_p(k) \leq -\eta_P.$$

(c) (Uniform principal-mode decomposition of fiber kernels) Let $p_{p,k}(t; \xi, \eta)$ be the kernel of $e^{t\mathcal{A}_p(k)}$. There exist functions $a_p(k; \xi, \eta)$ continuous on $P \times [-\delta_P, \delta_P] \times \mathbb{T} \times \mathbb{T}$ and constants $c_P, C_P > 0$ such that for all $p \in P$, $|k| \leq \delta_P$, $t \geq t_P$ and all $\xi, \eta \in \mathbb{T}$,

$$(37) \quad p_{p,k}(t; \xi, \eta) = a_p(k; \xi, \eta) e^{t\lambda_p(k)} + r_{p,k}(t; \xi, \eta), \quad |r_{p,k}(t; \xi, \eta)| \leq C_P e^{-\eta_P t}.$$

Moreover, after possibly shrinking δ_P , the leading amplitude satisfies the uniform bound

$$(38) \quad c_P \leq \Re a_p(k; 0, \eta) \leq |a_p(k; 0, \eta)| \leq C_P \quad \text{for all } (p, k, \eta) \in P \times [-\delta_P, \delta_P] \times \mathbb{T}.$$

Furthermore, after possibly shrinking δ_P further, there exists $C_{A,P} > 0$ such that

$$(39) \quad \sup_{(p, \eta) \in P \times \mathbb{T}} |a_p(k; 0, \eta) - a_p(0; 0, \eta)| \leq C_{A,P} |k| \quad (|k| \leq \delta_P).$$

(d) (Uniform far- k kernel bound) There exist constants $C_P^{\text{far}} \geq 1$ and $\eta_P^{\text{far}} > 0$ such that for all $p \in P$, all $|k| \in [\delta_P, \pi]$, all $t \geq 1$, and all $\xi, \eta \in \mathbb{T}$,

$$(40) \quad |p_{p,k}(t; \xi, \eta)| \leq C_P^{\text{far}} e^{-\eta_P^{\text{far}} t}.$$

Proof. Because P is compact and $(p, k) \mapsto \mathcal{A}_p(k)$ is continuous in operator topology on $H^2(\mathbb{T}) \rightarrow L^2(\mathbb{T})$, standard analytic perturbation theory applies uniformly on compact parameter sets.

Lemma (Kernel regularity on compact parameter sets [9, Chap. VII, §1], [14, Chap. 4, Sec. 1]). *Fix $t > 0$ and a compact set $P \subset \mathbb{R}$. Then for every $(p, k) \in P \times [-\pi, \pi]$ the fiber semigroup $e^{t\mathcal{A}_p(k)}$ on $C(\mathbb{T}; \mathbb{C})$ admits an integral kernel $p_{p,k}(t; \xi, \eta)$ which is continuous in $(\xi, \eta) \in \mathbb{T} \times \mathbb{T}$. Moreover, for each fixed $t > 0$ the map*

$$(p, k, \xi, \eta) \longmapsto p_{p,k}(t; \xi, \eta)$$

is jointly continuous on $P \times [-\pi, \pi] \times \mathbb{T} \times \mathbb{T}$. In particular,

$$\sup_{(p, k) \in P \times [-\pi, \pi]} \sup_{\xi, \eta \in \mathbb{T}} |p_{p,k}(t; \xi, \eta)| < \infty,$$

and $(p, k) \mapsto e^{t\mathcal{A}_p(k)}$ is continuous in operator norm $\|\cdot\|_{C(\mathbb{T}) \rightarrow C(\mathbb{T})}$.

Sketch of Proof. For each fixed (p, k) , $\mathcal{A}_p(k)$ is uniformly elliptic on the compact manifold $\mathbb{T}$ with C^1 coefficients. Standard parabolic regularity implies that $e^{t\mathcal{A}_p(k)}$ is smoothing and therefore admits a continuous kernel for every $t > 0$. On compact P , Lemma 4.7 together with the normalization $r_p(0) = 1$ implies that $p \mapsto r_p$ is continuous in $H^2(\mathbb{T})$, hence in $C^1(\mathbb{T})$. Using the eigenvalue equation

$$r_p'' = 2(H(p) + V - \tfrac{1}{2}p^2)r_p - 2pr_p',$$

we in fact obtain continuity of $p \mapsto r_p$ in $C^2(\mathbb{T})$. Therefore

$$b_p = p + \frac{r_p'}{r_p}$$

depends continuously on p in $C^1(\mathbb{T})$. Since the dependence on k is polynomial, the coefficients of $\mathcal{A}_p(k)$ depend continuously on (p, k) in C^1 , with uniformly bounded C^1 norms on $P \times [-\pi, \pi]$. Standard parabolic regularity then implies that the kernels depend continuously on (p, k) in the $C(\mathbb{T} \times \mathbb{T})$ topology; uniform boundedness follows by compactness. Finally, continuity in operator norm on $C(\mathbb{T}; \mathbb{C})$ follows from the integral representation. $\square$

Proof of (a). For each fixed p , Lemma 6.8 gives an expansion near $k = 0$. By compactness of P , there exist $\delta_0 \in (0, 1)$ and $C_0 \geq 1$ such that for all $p \in P$ and all $|k| \leq \delta_0$,

$$\lambda_p(k) = i m_p k - \frac{1}{2} \sigma_p^2 k^2 + R_p(k), \quad |R_p(k)| \leq C_0 |k|^3.$$

Moreover, by uniform analytic perturbation theory on compact parameter sets, the derivatives $\partial_k^j \lambda_p(0)$ ($j = 1, 2$) depend continuously on p . Since $\sigma_p^2 = -\lambda_p''(0)$, the map $p \mapsto \sigma_p^2$ is continuous on P . Hence

$$\sigma_P := \min_{p \in P} \sigma_p^2 > 0.$$

Proof of (c). For $|k| \leq \delta_0$, the principal eigenvalue $\lambda_p(k)$ is simple and uniformly isolated from the rest of the spectrum. Let $\Pi_{p,k}$ denote the corresponding Riesz projection. By Kato perturbation theory together with elliptic regularity, after possibly shrinking δ_0 to some $\delta_P \in (0, \delta_0]$, one can choose a right eigenfunction $u_{p,k} \in C^2(\mathbb{T})$ and a function $\psi_{p,k} \in C^2(\mathbb{T})$ representing the corresponding left eigenfunctional such that

$$\mathcal{A}_p(k)u_{p,k} = \lambda_p(k)u_{p,k}, \quad \int_{\mathbb{T}} \psi_{p,k}(\eta)u_{p,k}(\eta) d\eta = 1,$$

and

$$u_{p,0} \equiv 1, \quad \psi_{p,0} = \pi_p.$$

Since $u_{p,0}(0) = 1$, after shrinking δ_P if necessary we may further impose

$$u_{p,k}(0) = 1 \quad \text{for all } (p, k) \in P \times [-\delta_P, \delta_P].$$

With this gauge fixing, the maps

$$(p, k) \longmapsto u_{p,k}, \quad (p, k) \longmapsto \psi_{p,k}$$

are continuous on $P \times [-\delta_P, \delta_P]$ in $C^1(\mathbb{T})$, and analytic in k for each fixed p .

Define

$$a_p(k; \xi, \eta) := u_{p,k}(\xi) \psi_{p,k}(\eta).$$

Then $\Pi_{p,k}$ has kernel $a_p(k; \xi, \eta)$, and in particular

$$a_p(0; 0, \eta) = \pi_p(\eta).$$

Since $(p, \eta) \mapsto \pi_p(\eta)$ is continuous and strictly positive, after possibly shrinking δ_P further, there exists $c_P > 0$ such that

$$c_P \leq \Re a_p(k; 0, \eta) \leq |a_p(k; 0, \eta)| \quad \text{for all } (p, k, \eta) \in P \times [-\delta_P, \delta_P] \times \mathbb{T}.$$

By continuity on the same compact set, there exists $C_a < \infty$ such that

$$|a_p(k; 0, \eta)| \leq C_a \quad \text{for all } (p, k, \eta) \in P \times [-\delta_P, \delta_P] \times \mathbb{T}.$$

Hence (38) holds after replacing C_a by a larger constant if necessary.

For (39), note that for each fixed $p \in P$ the Riesz projection (hence the amplitude a_p) depends analytically on k for $|k| \leq \delta_P$. By Cauchy estimates on a complex- k disc of uniform radius, there exists $C_{A,P} > 0$ such that

$$\sup_{(p, \eta) \in P \times \mathbb{T}} |a_p(k; 0, \eta) - a_p(0; 0, \eta)| \leq C_{A,P} |k|, \quad |k| \leq \delta_P.$$

Let

$$S_{p,k}(t) := e^{tA_p(k)} - e^{t\lambda_p(k)} \Pi_{p,k}.$$

By uniform spectral separation of the principal branch from the rest of the spectrum on $P \times [-\delta_P, \delta_P]$, there exist constants $\eta_{\text{near}} > 0$ and $C_{\text{near}} \geq 1$ such that

$$\|S_{p,k}(t)\|_{C(\mathbb{T}) \rightarrow C(\mathbb{T})} \leq C_{\text{near}} e^{-\eta_{\text{near}} t}, \quad p \in P, \quad |k| \leq \delta_P, \quad t \geq 0.$$

Set $t_P := 1$. Applying the auxiliary lemma with $t = 1$ gives

$$M_1 := \sup_{p \in P, |k| \leq \delta_P, \xi, \eta \in \mathbb{T}} |p_{p,k}(1; \xi, \eta)| < \infty.$$

Since $\Pi_{p,k}$ commutes with $e^{tA_p(k)}$, for $t \geq 1$ we have

$$S_{p,k}(t) = S_{p,k}(t-1) e^{A_p(k)}.$$

Therefore, defining

$$r_{p,k}(t; \xi, \eta) := (S_{p,k}(t-1)[p_{p,k}(1; \cdot, \eta)])(\xi), \quad t \geq 1,$$

we obtain

$$p_{p,k}(t; \xi, \eta) = a_p(k; \xi, \eta) e^{t\lambda_p(k)} + r_{p,k}(t; \xi, \eta),$$

and

$$|r_{p,k}(t; \xi, \eta)| \leq C_{\text{near}} M_1 e^{-\eta_{\text{near}}(t-1)}.$$

Hence (37) holds for all $t \geq t_P$ after enlarging constants if necessary.

Now fix once and for all

$$C_P := \max\{1, C_0, C_a, C_{\text{near}} M_1 e^{\eta_{\text{near}}}\}.$$

Then part (a), the upper bound in (38), and (37) all hold with this same constant C_P .

Finally, after possibly shrinking δ_P once more, we may also assume

$$C_P \delta_P \leq \sigma_P / 8.$$

All previously established bounds remain valid on the smaller interval $|k| \leq \delta_P$. *Proof of (b).* Now use this final choice of δ_P . For each fixed p , Lemma 6.7 gives

$$\sup_{\delta_P \leq |k| \leq \pi} s(\mathcal{A}_p(k)) < 0.$$

By upper semicontinuity of $(p, k) \mapsto s(\mathcal{A}_p(k))$ on the compact set

$$P \times \{k : \delta_P \leq |k| \leq \pi\},$$

there exists $\eta_{\text{far}} > 0$ such that

$$s(\mathcal{A}_p(k)) \leq -\eta_{\text{far}}, \quad p \in P, \delta_P \leq |k| \leq \pi.$$

Set

$$\eta_P := \min\{\eta_{\text{near}}, \eta_{\text{far}}\} > 0.$$

Then both part (b) and the remainder bound in (37) hold with this common η_P .

Proof of (d). Fix $\tau := 1$ and set $T_{p,k} := e^{\tau \mathcal{A}_p(k)}$ acting on $C(\mathbb{T}; \mathbb{C})$ for $\delta_P \leq |k| \leq \pi$. By part (b),

$$r(T_{p,k}) = e^{\tau s(\mathcal{A}_p(k))} \leq e^{-\eta_P} < 1.$$

Choose $\rho \in (e^{-\eta_P}, 1)$. Then for every such (p, k) , the circle

$$\Gamma := \{z : |z| = \rho\}$$

lies in the resolvent set of $T_{p,k}$. By the auxiliary lemma with $t = \tau$, $(p, k) \mapsto T_{p,k}$ is continuous in operator norm, so on the compact parameter set

$$P \times \{k : \delta_P \leq |k| \leq \pi\}$$

we have a uniform resolvent bound

$$M_\rho := \sup_{(p,k)} \sup_{|z|=\rho} \|(zI - T_{p,k})^{-1}\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} < \infty.$$

By standard semigroup bounds for uniformly parabolic operators whose coefficients are uniformly bounded on the compact parameter set,

$$\sup_{(p,k), s \in [0,1]} \|e^{s \mathcal{A}_p(k)}\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} < \infty.$$

By Dunford functional calculus,

$$T_{p,k}^n = \frac{1}{2\pi i} \int_{\Gamma} z^n (zI - T_{p,k})^{-1} dz,$$

hence

$$\|T_{p,k}^n\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} \leq M_\rho \rho^{n+1}$$

uniformly in (p, k) . Writing $t = n + s$ with $n = \lfloor t \rfloor$ and $s \in [0, 1]$,

$$\|e^{t \mathcal{A}_p(k)}\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} = \|e^{s \mathcal{A}_p(k)} T_{p,k}^n\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} \leq \left(\sup_{(p,k), s \in [0,1]} \|e^{s \mathcal{A}_p(k)}\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} \right) M_\rho \rho^{n+1} \leq C_0^{\text{far}} e^{-\eta_P t}$$

for some constants $C_0^{\text{far}} \geq 1$ and $\eta_P^{\text{far}} > 0$.

Applying the auxiliary lemma again with $t = 1$ yields

$$M_1^{\text{far}} := \sup_{p \in P, \delta_P \leq |k| \leq \pi, \xi, \eta \in \mathbb{T}} |p_{p,k}(1; \xi, \eta)| < \infty.$$

For $t \geq 1$,

$$p_{p,k}(t; \xi, \eta) = (e^{(t-1)\mathcal{A}_p(k)}[p_{p,k}(1; \cdot, \eta)])(\xi),$$

so

$$|p_{p,k}(t; \xi, \eta)| \leq \|e^{(t-1)\mathcal{A}_p(k)}\|_{C(\mathbb{T}; \mathbb{C}) \rightarrow C(\mathbb{T}; \mathbb{C})} \|p_{p,k}(1; \cdot, \eta)\|_\infty \leq C_0^{\text{far}} M_1^{\text{far}} e^{-\eta_P^{\text{far}}(t-1)}.$$

After enlarging constants if necessary, this gives (40). $\square$

6.5. Local limit bound along the ballistic ray.

Proposition 6.10 (Local limit bound, two-sided). *Fix a compact set $P \subset \mathbb{R}$. There exist $t_0 \geq 1$ and constants $0 < c \leq C < \infty$ such that for every $p \in P$, every $t \geq t_0$, and every $r \in [-1, 1]$,*

$$(41) \quad \frac{c}{\sqrt{t}} \leq \tilde{p}_p(t, 0, m_p t + r) \leq \frac{C}{\sqrt{t}}.$$

Proof. Fix compact P and let

$$\delta_P, \sigma_P, \eta_P, c_P, C_P, C_{A,P}, C_P^{\text{far}}, \eta_P^{\text{far}}, t_P$$

be the constants furnished by Proposition 6.9. From now on assume

$$t \geq t_P.$$

At the end we choose $t_0 \geq t_P$ large enough so that all subsequent conditions are satisfied.

Write $y := m_p t + r$ and $z := y \bmod 1 \in \mathbb{T}$.

Step 1: Bloch representation specialized to $x = 0$. From Proposition 6.3 with $x = 0$,

$$(42) \quad \tilde{p}_p(t, 0, y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-iky} p_{p,k}(t; 0, z) dk.$$

Reality / conjugate symmetry. Since b_p is real-valued, we have the conjugation identity

$$\overline{\mathcal{A}_p(k)\phi} = \mathcal{A}_p(-k)\overline{\phi},$$

hence the corresponding kernels satisfy

$$p_{p,-k}(t; \xi, \eta) = \overline{p_{p,k}(t; \xi, \eta)}.$$

Consequently the integrand in (42) is conjugate-symmetric in k , and $\tilde{p}_p(t, 0, y)$ is real. In particular, for any symmetric interval $[-\alpha, \alpha] \subset [-\pi, \pi]$,

$$\int_{-\alpha}^{\alpha} e^{-iky} p_{p,k}(t; 0, z) dk = 2 \Re \int_0^{\alpha} e^{-iky} p_{p,k}(t; 0, z) dk.$$

Step 2: Split the k -integral and insert the principal-mode decomposition.

Split $[-\pi, \pi] = [-\delta_P, \delta_P] \cup \{|k| \in [\delta_P, \pi]\}$ and write

$$I_{\text{near}} := \frac{1}{2\pi} \int_{|k| \leq \delta_P} e^{-iky} p_{p,k}(t; 0, z) dk, \quad I_{\text{far}} := \frac{1}{2\pi} \int_{\delta_P \leq |k| \leq \pi} e^{-iky} p_{p,k}(t; 0, z) dk.$$

For I_{near} , use (37):

$$p_{p,k}(t; 0, z) = a_p(k; 0, z)e^{t\lambda_p(k)} + r_{p,k}(t; 0, z), \quad |r_{p,k}(t; 0, z)| \leq C_P e^{-\eta_P t}.$$

Hence

$$(43) \quad I_{\text{near}} = \frac{1}{2\pi} \int_{|k| \leq \delta_P} e^{-iky} a_p(k; 0, z) e^{t\lambda_p(k)} dk + \text{Err}_1,$$

where $|\text{Err}_1| \leq C_P e^{-\eta_P t}$.

For I_{far} , apply Proposition 6.9(d):

$$|I_{\text{far}}| \leq \frac{1}{2\pi} \int_{\delta_P \leq |k| \leq \pi} |p_{p,k}(t; 0, z)| dk \leq C_P^{\text{far}} e^{-\eta_P^{\text{far}} t}.$$

Combining (42) and (43), we reduce to estimating the main integral

$$(44) \quad J := \frac{1}{2\pi} \int_{|k| \leq \delta_P} e^{-iky} a_p(k; 0, z) e^{t\lambda_p(k)} dk.$$

If we set

$$c := \min\{\eta_P, \eta_P^{\text{far}}\},$$

then all remaining errors are $O(e^{-ct})$, hence $o(t^{-1/2})$ as $t \rightarrow \infty$.

Moreover, since $\tilde{p}_p(t, 0, y)$ is real-valued and

$$\tilde{p}_p(t, 0, y) = J + O(e^{-ct}),$$

we also have

$$(45) \quad J = \Re J + O(e^{-ct}).$$

Step 3: Center at the ballistic speed m_p and use the Taylor expansion. Write $y = m_p t + r$. Then

$$e^{-iky} e^{t\lambda_p(k)} = e^{-ikr} \exp(t(\lambda_p(k) - im_p k)).$$

By Proposition 6.9(a),

$$\lambda_p(k) - im_p k = -\frac{1}{2}\sigma_p^2 k^2 + R_p(k), \quad |R_p(k)| \leq C_P |k|^3, \quad \sigma_p^2 \geq \sigma_P.$$

Thus for $|k| \leq \delta_P$,

$$(46) \quad |\exp(t(\lambda_p(k) - im_p k))| = \exp(t\Re(\lambda_p(k) - im_p k)) \leq \exp\left(-\frac{1}{2}\sigma_P t k^2 + C_P t |k|^3\right).$$

Step 4: Upper bound. We bound the main integral J in (44). By (38) we have $|a_p(k; 0, z)| \leq C_P$ for $|k| \leq \delta_P$. By the final choice of δ_P in Proposition 6.9, we have

$$C_P |k| \leq \sigma_P / 8 \quad \text{for all } |k| \leq \delta_P.$$

Then for $|k| \leq \delta_P$,

$$t\Re(\lambda_p(k) - im_p k) \leq -\frac{1}{2}\sigma_P t k^2 + t|R_p(k)| \leq -\frac{1}{2}\sigma_P t k^2 + C_P t |k|^3 \leq -\frac{3}{8}\sigma_P t k^2.$$

Hence

$$|J| \leq \frac{C_P}{2\pi} \int_{|k| \leq \delta_P} e^{-\frac{3}{8}\sigma_P t k^2} dk \leq \frac{C}{\sqrt{t}}.$$

Adding the exponentially small errors gives the upper bound in (41).

Step 5: Lower bound via a $K/\sqrt{t}$ window (revised). Fix $K \geq 1$ (to be chosen below). Decompose

$$J = J_{\leq K} + J_{> K},$$

where

$$J_{\leq K} := \frac{1}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-iky} a_p(k; 0, z) e^{t\lambda_p(k)} dk, \quad J_{> K} := \frac{1}{2\pi} \int_{K/\sqrt{t} \leq |k| \leq \delta_P} e^{-iky} a_p(k; 0, z) e^{t\lambda_p(k)} dk.$$

Assume t is large so that $K/\sqrt{t} \leq \delta_P$ and $K/\sqrt{t} \leq \pi/3$ (then $|kr| \leq \pi/3$ since $|r| \leq 1$). Write $y = m_p t + r$ as before, and rewrite

$$J_{\leq K} = \frac{1}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-ikr} a_p(k; 0, z) e^{-\frac{1}{2}\sigma_p^2 t k^2} e^{tR_p(k)} dk.$$

Let $a_0 := a_p(0; 0, z) = \pi_p(z)$, which is real and strictly positive. Set

$$\mathcal{E}_t(k) := a_p(k; 0, z) e^{tR_p(k)} - a_0.$$

On $|k| \leq K/\sqrt{t}$, by (39) and $|R_p(k)| \leq C_P |k|^3$ we have

$$|a_p(k; 0, z) - a_0| \leq C_{A,P} |k| \leq \frac{C_{A,P} K}{\sqrt{t}}, \quad |e^{tR_p(k)} - 1| \leq e^{|tR_p(k)|} |tR_p(k)| \leq \frac{C_K}{\sqrt{t}},$$

hence $|\mathcal{E}_t(k)| \leq C_K/\sqrt{t}$ on this window (for a constant C_K depending only on K and the uniform constants).

Therefore

$$J_{\leq K} = \frac{a_0}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-ikr} e^{-\frac{1}{2}\sigma_p^2 t k^2} dk + \frac{1}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-ikr} e^{-\frac{1}{2}\sigma_p^2 t k^2} \mathcal{E}_t(k) dk.$$

Taking real parts and using symmetry (the sine part cancels),

$$\begin{aligned} \Re J_{\leq K} &= \frac{a_0}{2\pi} \int_{|k| \leq K/\sqrt{t}} \cos(kr) e^{-\frac{1}{2}\sigma_p^2 t k^2} dk + \frac{1}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-\frac{1}{2}\sigma_p^2 t k^2} \Re(e^{-ikr} \mathcal{E}_t(k)) dk \\ &\geq \frac{a_0}{2\pi} \int_{|k| \leq K/\sqrt{t}} \cos(kr) e^{-\frac{1}{2}\sigma_p^2 t k^2} dk - \frac{1}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-\frac{1}{2}\sigma_p^2 t k^2} |\mathcal{E}_t(k)| dk. \end{aligned}$$

For the main term, since $|kr| \leq K/\sqrt{t} \leq \pi/3$, we have $\cos(kr) \geq 1/2$. Also set

$$\bar{\sigma}_P^2 := \max_{p \in P} \sigma_p^2 < \infty$$

(by continuity on compact P). Then $e^{-\frac{1}{2}\sigma_p^2 t k^2} \geq e^{-\frac{1}{2}\bar{\sigma}_P^2 t k^2}$, so

$$\frac{a_0}{2\pi} \int_{|k| \leq K/\sqrt{t}} \cos(kr) e^{-\frac{1}{2}\sigma_p^2 t k^2} dk \geq \frac{a_0}{4\pi} \int_{|k| \leq K/\sqrt{t}} e^{-\frac{1}{2}\bar{\sigma}_P^2 t k^2} dk = \frac{a_0}{4\pi\sqrt{t}} \int_{|u| \leq K} e^{-\frac{1}{2}\bar{\sigma}_P^2 u^2} du.$$

Moreover $a_0 = \pi_p(z) \geq c_P$ by (38) at $k = 0$, hence

$$\frac{a_0}{4\pi\sqrt{t}} \int_{|u| \leq K} e^{-\frac{1}{2}\sigma_P^2 u^2} du \geq \frac{c_P}{4\pi\sqrt{t}} \int_{|u| \leq K} e^{-\frac{1}{2}\sigma_P^2 u^2} du =: \frac{c_1(K)}{\sqrt{t}}, \quad c_1(K) > 0.$$

For the error term, using $|\mathcal{E}_t(k)| \leq C_K/\sqrt{t}$ and $e^{-\frac{1}{2}\sigma_P^2 tk^2} \leq e^{-\frac{1}{2}\sigma_P tk^2}$ (since $\sigma_P^2 \geq \sigma_P$),

$$\frac{1}{2\pi} \int_{|k| \leq K/\sqrt{t}} e^{-\frac{1}{2}\sigma_P^2 tk^2} |\mathcal{E}_t(k)| dk \leq \frac{C_K}{2\pi\sqrt{t}} \int_{|k| \leq K/\sqrt{t}} e^{-\frac{1}{2}\sigma_P tk^2} dk = \frac{C'_K}{t}.$$

Choosing t large so that $C'_K/t \leq \frac{1}{2}c_1(K)/\sqrt{t}$, we obtain

$$\Re J_{\leq K} \geq \frac{c_1(K)}{2\sqrt{t}}.$$

Next, bound the tail $J_{>K}$ using Step 4: there exists $c_{\text{dec}} > 0$ such that for $|k| \leq \delta_P$, $t\Re(\lambda_p(k) - im_pk) \leq -c_{\text{dec}}tk^2$, hence

$$|J_{>K}| \leq \frac{C}{2\pi} \int_{K/\sqrt{t} \leq |k| \leq \delta_P} e^{-c_{\text{dec}}tk^2} dk \leq \frac{C}{\sqrt{t}} \int_{|u| \geq K} e^{-c_{\text{dec}}u^2} du =: \frac{c_2(K)}{\sqrt{t}}.$$

Choose K so large that $c_2(K) \leq \frac{1}{4}c_1(K)$ (Gaussian tail $\rightarrow 0$).

Then for t large (so that all window conditions hold and the exponentially small errors are $o(t^{-1/2})$),

$$\tilde{p}_p(t, 0, m_pt + r) = J + O(e^{-ct}) = \Re J + O(e^{-ct}) \quad \text{by (45),}$$

and hence

$$\tilde{p}_p(t, 0, m_pt + r) = \Re J_{\leq K} + \Re J_{>K} + O(e^{-ct}) \geq \Re J_{\leq K} - |J_{>K}| - O(e^{-ct}) \geq \frac{c_1(K)}{4\sqrt{t}}.$$

This yields the lower bound in (41) and completes the proof. $\square$

7. BALLISTIC BOUNDS FOR THE SCHRÖDINGER KERNEL

We now transfer the local limit bound for $\tilde{p}_p$ back to K .

7.1. Comparing the tilted endpoint density and $\tilde{p}_p$. Define the tilted endpoint density

$$(47) \quad q_{t,p}(y) := \frac{e^{py}K(t; 0, y)}{\Psi_t(p)}.$$

Lemma 7.1 (Equivalence $q_{t,p} \asymp \tilde{p}_p$ for large t). *Fix compact $P \subset \mathbb{R}$. There exist $t_0 \geq 1$ and $0 < c \leq C$ such that for all $p \in P$, $t \geq t_0$, and all $y \in \mathbb{R}$,*

$$(48) \quad c\tilde{p}_p(t, 0, y) \leq q_{t,p}(y) \leq C\tilde{p}_p(t, 0, y).$$

Proof. From Lemma 5.4 with $x = 0$ and $h_p(y) = e^{py}r_p(y \bmod 1)$,

$$e^{py}K(t; 0, y) = e^{tH(p)}r_p(0) \frac{1}{r_p(y \bmod 1)} \tilde{p}_p(t, 0, y).$$

Divide by $\Psi_t(p)$ and use Corollary 4.5 to bound $\Psi_t(p)$ by constants times $e^{tH(p)}$ uniformly on compact P , and use $\min_{\mathbb{T}} r_p > 0$ and $\max_{\mathbb{T}} r_p < \infty$ uniformly on compact P (continuity in p and compactness under any fixed continuous normalization). $\square$

7.2. Ballistic estimate. For $p \in \mathbb{R}$, set $v := m_p = H'(p)$ (Lemma 6.2).

Proposition 7.2 (Ballistic kernel bound on compact velocity sets). *Let $V_0 \subset \mathbb{R}$ be compact. Then there exist $t_0 \geq 1$ and $0 < c \leq C < \infty$ such that: for every $v \in V_0$, if $p = p(v)$ is chosen so that $H'(p) = v$, then for all $t \geq t_0$ and all $r \in [-1, 1]$,*

$$(49) \quad \frac{c}{\sqrt{t}} e^{-tL(v)} \leq K(t; 0, vt + r) \leq \frac{C}{\sqrt{t}} e^{-tL(v)}.$$

Proof. Fix compact V_0 and let $P_0 := \{p(v) : v \in V_0\}$.

By Lemma 4.8, H is convex; by Lemma 4.7, H is C^1 and H' is continuous and nondecreasing. Using convexity and (16), for $p > 0$,

$$H'(p) \geq \frac{H(p) - H(0)}{p} \geq \frac{\frac{1}{2}p^2 - \|V\|_\infty - H(0)}{p} = \frac{p}{2} - \frac{\|V\|_\infty + H(0)}{p} \xrightarrow{p \rightarrow \infty} \infty.$$

Similarly for $p < 0$, convexity implies $H'(p) \leq \frac{H(p) - H(0)}{p}$, and using the lower quadratic bound in (16) gives

$$H'(p) \leq \frac{H(p) - H(0)}{p} \leq \frac{\frac{1}{2}p^2 - \|V\|_\infty - H(0)}{p} = \frac{p}{2} - \frac{\|V\|_\infty + H(0)}{p} \xrightarrow{p \rightarrow -\infty} -\infty.$$

Hence $H' : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and onto. Since H is convex, H' is nondecreasing. If H' were constant on some nontrivial interval, then H would be affine on that interval; by analyticity this would force H to be affine on all of $\mathbb{R}$, contradicting the quadratic growth (16). Therefore H' is strictly increasing, so $p(v)$ is uniquely defined and depends continuously on v . Therefore P_0 is compact.

For $v = H'(p)$ and $y = vt + r$, we have by definition (47)

$$K(t; 0, y) = \Psi_t(p) e^{-py} q_{t,p}(y).$$

Using Corollary 4.5 uniformly on compact P_0 gives $\Psi_t(p) \asymp e^{tH(p)}$. Using Lemma 7.1 and Proposition 6.10 uniformly on P_0 yields $q_{t,p}(vt + r) \asymp t^{-1/2}$.

Therefore

$$K(t; 0, vt + r) \asymp \frac{1}{\sqrt{t}} \exp(tH(p) - p(vt + r)) = \frac{1}{\sqrt{t}} \exp(t(H(p) - pv)) \cdot e^{-pr}.$$

Absorb e^{-pr} into the constants since $r \in [-1, 1]$ and $p \in P_0$. Finally, standard convex duality gives $L(v) = pv - H(p)$ when $v = H'(p)$ (see [16, Thm. 23.5]), hence $H(p) - pv = -L(v)$, proving (49). $\square$

8. LAPLACE ASYMPTOTICS AND PROOF OF THEOREM 2.1

Recall $T = 2/\varepsilon$ and (10). Change variables $x = \varepsilon y$, i.e. $y = x/\varepsilon$ and $dy = dx/\varepsilon$:

$$(50) \quad u_\varepsilon(T, 0) = \frac{1}{\varepsilon} \int_{\mathbb{R}} K\left(\frac{2}{\varepsilon}; 0, \frac{x}{\varepsilon}\right) \exp\left(-\frac{g(x)}{2\varepsilon}\right) dx.$$

Define the Laplace-phase function

$$(51) \quad F(x) := \frac{g(x)}{2} + 2L\left(\frac{x}{2}\right).$$

By (17) and boundedness of g , F is coercive: $F(x) \rightarrow \infty$ as $|x| \rightarrow \infty$. Let

$$(52) \quad m := \inf_{x \in \mathbb{R}} F(x), \quad c_0 := -m,$$

matching (3).

8.1. Uniform kernel control on the relevant compact x -set. Choose $R \geq 1$ so that

$$(53) \quad \text{(i) the minimizer set } \arg \min F \subset [-R, R], \quad \text{and (ii) } \frac{R^2}{4} \geq m + 2 + 2\|V\|_\infty + \frac{1}{2}\|g\|_\infty.$$

Such an R exists because F is coercive.

Set $V_0 := [-R/2, R/2]$ for the velocity window. For $x \in [-R, R]$, define $v := x/2 \in V_0$.

By Proposition 7.2 (with $r = 0$ and $t = 2/\varepsilon$), there exist constants $0 < c_R \leq C_R < \infty$ and $\varepsilon_0 > 0$ such that for all $\varepsilon \in (0, \varepsilon_0)$ and all $x \in [-R, R]$,

$$(54) \quad c_R \sqrt{\varepsilon} \exp\left(-\frac{2}{\varepsilon}L(x/2)\right) \leq K\left(\frac{2}{\varepsilon}; 0, \frac{x}{\varepsilon}\right) \leq C_R \sqrt{\varepsilon} \exp\left(-\frac{2}{\varepsilon}L(x/2)\right).$$

8.2. Tail bound outside $[-R, R]$. We will use a crude but uniform Gaussian bound for K . From Lemma 3.2 and (9),

$$K(t; 0, y) \leq e^{\|V\|_\infty t} p_t(y).$$

With $t = 2/\varepsilon$ and $y = x/\varepsilon$,

$$K\left(\frac{2}{\varepsilon}; 0, \frac{x}{\varepsilon}\right) \leq C \sqrt{\varepsilon} \exp\left(\frac{2\|V\|_\infty}{\varepsilon} - \frac{x^2}{4\varepsilon}\right).$$

Since g is bounded, $\exp(-g(x)/(2\varepsilon)) \leq \exp(\|g\|_\infty/(2\varepsilon))$. Hence for $|x| > R$,

$$(55) \quad K\left(\frac{2}{\varepsilon}; 0, \frac{x}{\varepsilon}\right) \exp\left(-\frac{g(x)}{2\varepsilon}\right) \leq C \sqrt{\varepsilon} \exp\left(\frac{2\|V\|_\infty + \|g\|_\infty/2}{\varepsilon} - \frac{x^2}{4\varepsilon}\right).$$

Integrating (55) over $|x| > R$ and using the Gaussian tail bound $\int_{|x|>R} e^{-x^2/(4\varepsilon)} dx \leq C \sqrt{\varepsilon} e^{-R^2/(4\varepsilon)}$ gives

$$(56) \quad \frac{1}{\varepsilon} \int_{|x|>R} K\left(\frac{2}{\varepsilon}; 0, \frac{x}{\varepsilon}\right) \exp\left(-\frac{g(x)}{2\varepsilon}\right) dx \leq C \exp\left(\frac{2\|V\|_\infty + \|g\|_\infty/2 - R^2/4}{\varepsilon}\right).$$

By the choice (53)(ii), the exponent is $\leq -(m+2)/\varepsilon$, hence the tail is

$$(57) \quad \leq C e^{-(m+2)/\varepsilon}.$$

This is exponentially smaller than the scale $e^{-m/\varepsilon}$ that appears in the main contribution below; hence, once the lower bound on the main term is established, it is negligible in the logarithmic asymptotics.

8.3. Reduction to a Laplace integral on $[-R, R]$. Split (50) into $|x| \leq R$ and $|x| > R$. Using (54) for $|x| \leq R$ and (57) for $|x| > R$, we obtain for ε small:

$$(58) \quad c_R \varepsilon^{-1/2} \int_{-R}^R \exp\left(-\frac{F(x)}{\varepsilon}\right) dx \leq u_\varepsilon(T, 0) \leq C_R \varepsilon^{-1/2} \int_{-R}^R \exp\left(-\frac{F(x)}{\varepsilon}\right) dx + C e^{-(m+2)/\varepsilon}.$$

To pass from $[-R, R]$ to all of $\mathbb{R}$, note that by (17),

$$F(x) \geq \frac{x^2}{4} - 2\|V\|_\infty - \frac{1}{2}\|g\|_\infty.$$

Hence, by the choice of R ,

$$F(x) - m \geq \frac{x^2 - R^2}{4} + 2, \quad |x| \geq R.$$

Therefore

$$\int_{|x|>R} \exp\left(-\frac{F(x)}{\varepsilon}\right) dx \leq e^{-m/\varepsilon} e^{-2/\varepsilon} \int_{|x|>R} \exp\left(-\frac{x^2 - R^2}{4\varepsilon}\right) dx \leq C\sqrt{\varepsilon} e^{-(m+2)/\varepsilon}.$$

On the other hand, let $x_* \in [-R, R]$ be a minimizer of F . Since g is Lipschitz and L is convex, the function

$$F(x) = \frac{g(x)}{2} + 2L(x/2)$$

is locally Lipschitz on $\mathbb{R}$ (cf. Lemma 8.1 below for the convex part). Hence there exist $\delta, M > 0$ such that

$$0 \leq F(x) - m \leq M|x - x_*| \quad (|x - x_*| \leq \delta).$$

For all sufficiently small ε , one of the intervals

$$[x_* - \varepsilon, x_*] \quad \text{or} \quad [x_*, x_* + \varepsilon]$$

is contained in $[-R, R]$; denote it by I_ε . Then

$$\int_{-R}^R \exp\left(-\frac{F(x)}{\varepsilon}\right) dx \geq \int_{I_\varepsilon} \exp\left(-\frac{F(x)}{\varepsilon}\right) dx \geq e^{-m/\varepsilon} \int_0^\varepsilon e^{-Ms/\varepsilon} ds \geq c\varepsilon e^{-m/\varepsilon}.$$

Thus

$$\int_{\mathbb{R}} \exp\left(-\frac{F(x)}{\varepsilon}\right) dx \asymp \int_{-R}^R \exp\left(-\frac{F(x)}{\varepsilon}\right) dx,$$

and therefore

$$(59) \quad u_\varepsilon(T, 0) = \Theta\left(\varepsilon^{-1/2} \int_{\mathbb{R}} \exp\left(-\frac{F(x)}{\varepsilon}\right) dx\right),$$

where $\Theta(\cdot)$ means bounded above and below by positive constants independent of ε .

Define

$$(60) \quad J_\varepsilon := \int_{\mathbb{R}} \exp\left(-\frac{F(x) - m}{\varepsilon}\right) dx,$$

so that

$$\int_{\mathbb{R}} e^{-F(x)/\varepsilon} dx = e^{-m/\varepsilon} J_\varepsilon.$$

Thus (59) becomes

$$(61) \quad u_\varepsilon(T, 0) = \Theta(\varepsilon^{-1/2} e^{-m/\varepsilon} J_\varepsilon).$$

Taking $\varepsilon \log$ yields

$$(62) \quad c(\varepsilon) = \varepsilon \log u_\varepsilon(T, 0) = -m + \varepsilon \log(\varepsilon^{-1/2} J_\varepsilon) + O(\varepsilon).$$

8.4. Pure Laplace bounds: $J_\varepsilon \in [c\varepsilon, C]$.

Lemma 8.1 (Local Lipschitz of convex functions). *If $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is convex, then φ is Lipschitz on every compact interval.*

Proof. For convex φ , slopes are monotone:

$$\frac{\varphi(x) - \varphi(y)}{x - y} \leq \frac{\varphi(u) - \varphi(v)}{u - v} \quad \text{whenever } x < y \leq v < u.$$

Fix a compact interval $[a, b]$ and compare slopes to those at endpoints to bound the local Lipschitz constant. $\square$

Lemma 8.2 (Lower bound $J_\varepsilon \geq c\varepsilon$). *There exists $c > 0$ such that for ε sufficiently small,*

$$(63) \quad J_\varepsilon \geq c\varepsilon.$$

Proof. Let x_* be a minimizer of F so that $F(x_*) = m$. Since g is Lipschitz and L is convex, Lemma 8.1 implies F is Lipschitz on $[x_* - 1, x_* + 1]$: there exists $M > 0$ such that $0 \leq F(x) - m \leq M|x - x_*|$ for $|x - x_*| \leq 1$. Then

$$J_\varepsilon \geq \int_{|x - x_*| \leq \varepsilon} e^{-M|x - x_*|/\varepsilon} dx = 2 \int_0^\varepsilon e^{-Mt/\varepsilon} dt \geq c\varepsilon.$$

$\square$

Lemma 8.3 (Upper bound $J_\varepsilon \leq C$). *There exists $C > 0$ such that for all sufficiently small ε ,*

$$J_\varepsilon \leq C.$$

Proof. Recall $F(x) = \frac{g(x)}{2} + 2L(x/2)$ and $m = \inf_{\mathbb{R}} F$. By (17) we have, for all $x \in \mathbb{R}$,

$$L(x/2) \geq \frac{(x/2)^2}{2} - \|V\|_\infty = \frac{x^2}{8} - \|V\|_\infty.$$

Since g is bounded below by $-\|g\|_\infty$, it follows that

$$F(x) \geq -\frac{1}{2}\|g\|_\infty + 2\left(\frac{x^2}{8} - \|V\|_\infty\right) = \frac{x^2}{4} - 2\|V\|_\infty - \frac{1}{2}\|g\|_\infty.$$

Hence there exists $C_0 \in \mathbb{R}$ such that

$$F(x) - m \geq \frac{x^2}{4} - C_0 \quad \text{for all } x \in \mathbb{R}.$$

Choose $R \geq 1$ such that $\frac{x^2}{8} \geq C_0$ for all $|x| \geq R$. Then for $|x| \geq R$,

$$F(x) - m \geq \frac{x^2}{8}.$$

Therefore,

$$J_\varepsilon = \int_{\mathbb{R}} \exp\left(-\frac{F(x) - m}{\varepsilon}\right) dx \leq \int_{|x| \leq R} 1 dx + \int_{|x| > R} \exp\left(-\frac{x^2}{8\varepsilon}\right) dx \leq 2R + C\sqrt{\varepsilon}.$$

Since $\sqrt{\varepsilon} \leq 1$ for $\varepsilon \leq 1$, we get $J_\varepsilon \leq 2R + C =: C_1$, uniformly in small ε . $\square$

Completion of the proof of Theorem 2.1. Combining Lemmas 8.2 and 8.3,

$$(64) \quad c\varepsilon \leq J_\varepsilon \leq C.$$

Therefore

$$\varepsilon \log(\varepsilon^{-1/2} J_\varepsilon) = \varepsilon \left(-\frac{1}{2} \log \varepsilon + \log J_\varepsilon \right)$$

is bounded in absolute value by $C\varepsilon|\log \varepsilon|$ using (64). Insert into (62) and recall $c_0 = -m$ to obtain (4). $\square$

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